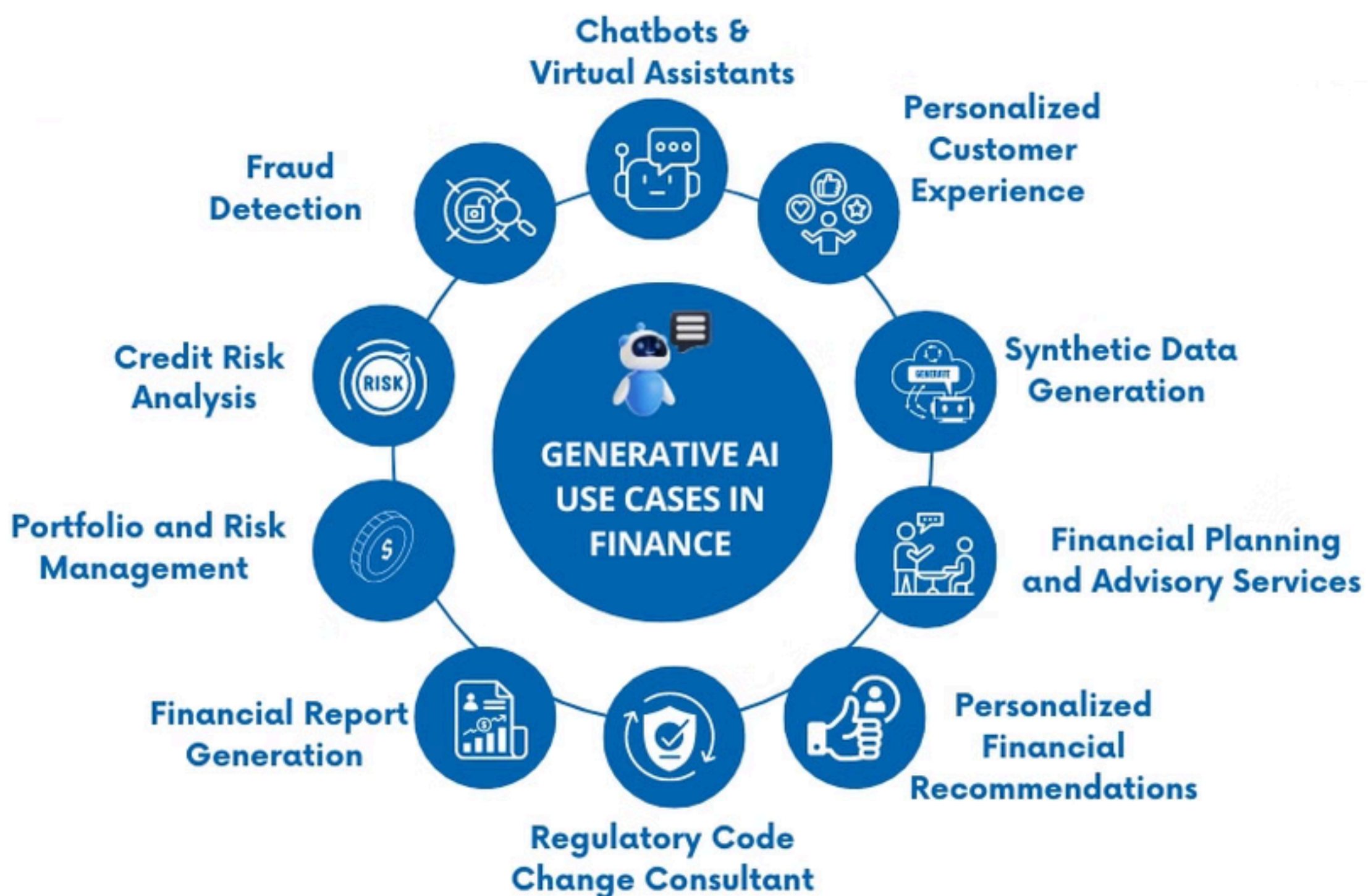


Day 28 of Agentic AI

Advanced Projects in Financial Analysis and AI Assistants



AI is rewriting how financial decisions get made. In this track, you build intelligent assistants and analytical systems that don't just crunch numbers – they understand context, learn from patterns, and guide decisions like a seasoned analyst. You'll move beyond dashboards and static models. The goal is to combine market intuition, statistical rigor, and modern AI capabilities to solve real investment and corporate finance problems.

Core Themes

Market-aware intelligence: Models that interpret market signals, not just historical datasets. Think: market sentiment parsing, risk flags, and real-time triggers.

Explainable decision support: Tools that can justify recommendations. If the system predicts a debt default or flags an overvalued asset, it must explain why.

AI-powered financial research: Assistants that can read earnings calls, understand reports, and extract insights faster than any analyst.

Flagship Projects

1. **Equity Research Assistant with Natural Language**

Understanding: Build an agent that digests earnings transcripts, analyst notes, and regulatory filings, then produces a research brief with key drivers, red flags, and valuation sanity checks.

Skills: LLMs, RAG frameworks, financial text processing, valuation frameworks.

2. Portfolio Optimization Engine with Risk Scenarios: Go beyond mean-variance optimization. Incorporate regime shifts, factor sensitivities, and downside-risk metrics.

Skills: Portfolio theory, Monte Carlo simulations, machine learning for scenario modeling.

3. Credit Default Prediction with Explainable AI: Train models to predict credit deterioration for corporate borrowers or retail customers, with interpretable outputs.

Skills: Classification models, SHAP/LIME, credit scoring frameworks, macroeconomic overlays.

4. Cash-Flow Intelligence Assistant for CFO Teams

A conversational planning tool that forecasts cash flows, stress-tests working capital, and helps CFOs model growth vs. liquidity trade-offs.

Skills: Time-series modeling, generative AI for financial planning, probabilistic forecasting.

5. Fraud Detection and Transaction Risk Agent

Use anomaly detection and graph analytics to flag suspect activity and explain risk vectors.

Skills: Graph ML, clustering, feature engineering for transactional data.

What You'll Learn

- Blend quantitative models with language-driven intelligence
- Design financial systems that communicate insights clearly
- Build AI agents that automate research and reasoning, not just calculations
- Apply risk concepts and performance metrics in real-world scenarios
- Work with noisy market data and corporate disclosures